

ESAT Mathematics 2

SPEED, RECALL & PROBLEM-SOLVING GUIDE

Speed, Recall & Problem-Solving Guide

Revised edition — real typeset formulas, a fixed 3-tier priority system,
and additional speed tricks folded in

Basis and scope

Syllabus content and terminology are based on UAT-UK, Notes on Mathematics for TMUA Paper 1 and ESAT Mathematics 2, June 2026. The official notes stress that the assessment is designed to test application of accessible mathematical knowledge to unfamiliar problems; most of the work should be in the thinking rather than long computation. This guide therefore prioritises recognition, method choice and exact manipulation.

Priority rankings, shortcuts and speed tactics are strategic synthesis. They are not official topic weightings or claims about question frequency.

How to use this guide

Revision note: the priority tiers below are relabelled from the original P1/P2/P3 numbering to AUTOMATE / FLUENT / TRAP-GUARD. The original numbering collided with the syllabus's own unit codes (e.g. Physics unit "P3 Mechanics" vs. priority tag "P1" for that same row) — the new labels mean the same thing but can never be confused with a unit code.

AUTOMATE. Instant recognition, zero hesitation.

FLUENT. A few written steps, but no uncertainty about method.

TRAP-GUARD. Familiar content where avoidable marks are lost.

- Memorise cold. Facts, identities and patterns that should not consume thinking time.
- Speed trick. A valid shortcut or method-selection rule; use it only when its conditions are satisfied.

Overall preparation priorities

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	MM1 Algebra & functions	The common language of the entire paper: structure, factorisation, functions, inequalities and quadratics.
AUTOMATE	MM8 Graphs of functions	Turns algebra into geometry; transformations, roots, intersections and shape can replace long manipulation.
AUTOMATE	MM4 Trigonometry	Exact values, rule selection, ambiguity and complete solution sets require very fast pattern recognition.
AUTOMATE	MM3 Coordinate geometry	Lines, circles and tangency connect algebra and geometry in a compact problem-solving toolkit.
FLUENT	MM6 Differentiation	Mechanically simple; speed comes from simplifying first and interpreting derivative information correctly.
FLUENT	MM7 Integration	Again technically narrow; signed area, symmetry and limits are the main conceptual traps.
FLUENT	MM2 Sequences & series	Formula recall is easy; indexing, recurrence cycles and binomial term selection are the main time sinks.
FLUENT	MM5 Exponentials & logarithms	Small syllabus but algebraically rich; hidden quadratics and log domains matter.

Each unit below follows the same shape: Priority ranking, Memorise cold, Ranked speed tricks (plus any additional tricks folded in for this revision, clearly marked), Common time-wasters and traps, and a 10-second pre-answer check.

MM1. Algebra and functions

Official guide: pp. 6-46 | Sections MM1.1-MM1.7

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

See structure before manipulation. This unit is not seven separate topics: indices, surds, quadratics, inequalities, polynomials and functions repeatedly reduce to factorisation, domain control and choosing the representation that preserves information.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Quadratics and discriminant	Factorisation, completed square, formula and graph interpretation are interchangeable tools.
AUTOMATE	Inequalities and modulus	Unsafe multiplication/division and lost solution regions are common traps.
AUTOMATE	Polynomial factorisation/division	Factor and remainder theorems can collapse apparently long algebra.
AUTOMATE	Surds and exact forms	Rationalisation and conjugates should be almost mechanical.
FLUENT	Simultaneous equations	Line plus quadratic reduces to one quadratic; the discriminant also gives intersection count.
FLUENT	Functions and domains	One-to-one/many-to-one, sqrt and modulus require precise interpretation.
FLUENT	Index laws	Should be automatic; the difficulty is usually recognising a disguised common base.

MEMORISE COLD

- Indices (positive base in the guide's treatment): $a^m a^n = a^{m+n}$; $a^m/a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^{-m} = 1/a^m$; $a^{1/n} = \text{nth root of } a$; $a^0 = 1$.
- Surds: extract square factors; rationalise a single-surd denominator by multiplying by the surd, and a binomial surd denominator by its conjugate.
- Difference of squares: $A^2 - B^2 = (A-B)(A+B)$. Sum/difference of cubes are useful: $A^3 - B^3 = (A-B)(A^2 + AB + B^2)$; $A^3 + B^3 = (A+B)(A^2 - AB + B^2)$.
- Quadratic formula: $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$. Discriminant $D = b^2 - 4ac$: $D > 0$ two real roots; $D = 0$ repeated root; $D < 0$ no real roots.
- Vertex x-coordinate of $ax^2 + bx + c$ is $-b/(2a)$. Completed square links the algebra directly to graph translation and the min/max.
- Factor theorem: $(x-a)$ is a factor of $f(x)$ iff $f(a) = 0$. Remainder on division by $(x-a)$ is $f(a)$.
- Function: a rule plus domain with exactly one output for each input. $\sqrt{x}$ denotes the non-negative root; $|x-a|$ is distance from a on the number line.

RANKED SPEED TRICKS

- Factorise before invoking machinery. For quadratics and cubics, test obvious factors/roots first. Formula and long division are fallbacks, not defaults.
- Use the discriminant as information, not just arithmetic. It tells you root count, line-quadratic intersections and tangency without solving for the roots.

3. Complete the square when the question is really about shape. Vertex, minimum/maximum, translations and root existence become visible at once.
4. Treat surds like algebraic symbols until the final simplification. When expanding $(a+b\sqrt{c})^2$, expand first, simplify roots last.
5. For $\sqrt{A+B\sqrt{C}}$, work backwards from $(p+q\sqrt{C})^2$. The middle term gives $2pq$ and the rational part gives p^2+Cq^2 ; try factor pairs rather than random guessing.
6. Inequality with an unknown-sign denominator: do not multiply by it blindly. The official notes show that multiplying by its square can preserve the inequality direction (while separately respecting points where the original denominator is zero).
7. Modulus means distance. $|x-a| < |x-b|$ often means "closer to a than b"; the midpoint can solve it almost instantly.
8. Line plus quadratic: substitute the line. You get one quadratic. Factor/discriminant then tells you two intersections, tangency or none.
9. Polynomial division: insert missing powers as zero terms. This prevents column-shift errors in long division.
10. Factor theorem root hunt: use structure. For integer-coefficient polynomials, factors of the constant term are the fastest sensible candidates for simple integer roots.
11. Do not prove an identity by examples. Substitution can disprove or check; proof requires algebra valid across the domain.
12. Check domain at the end. Square roots, denominators and later logarithms can make an algebraic candidate invalid.

OPTIONAL EXTRAS — *useful if already familiar, not a preparation priority*

- The Rational Root Theorem: possible rational roots of a polynomial are $\pm(\text{factor of constant})/(\text{factor of leading coefficient})$. Turns factor-theorem guessing into a short, checkable list. Valid mathematics, but beyond what the official notes ask you to know — useful if already familiar, not something to learn specifically for this test.
- Vieta's formulas: for $a \cdot x^2 + b \cdot x + c = 0$, sum of roots = $-b/a$, product of roots = c/a — a quick way to check a factorisation without solving. As above: fine if already known, not a preparation priority.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using the quadratic formula before checking an easy factorisation.
- Multiplying an inequality by an expression of unknown sign.
- Forgetting that $\sqrt{a}$ denotes the positive square root, while solving $x^2=a$ may give $\pm\sqrt{a}$.
- Cancelling across addition, e.g. treating $(x+2)/x$ as though the x cancels.
- Treating a few numerical substitutions as proof of an identity.
- Forgetting excluded values after simplifying a rational expression.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Factorise?
- Graph/completed square/discriminant more informative?
- Any denominator sign issue?
- Any domain exclusions?
- Modulus as distance?
- Could a simple candidate root collapse the polynomial?

EXPANDED PATTERN BANK

Mini Speed Drill

- Rationalise $3/(2+\sqrt{5}) \rightarrow$ multiply by $(\sqrt{5}-2)/(\sqrt{5}-2)$ to get $3\sqrt{5}-6$.
- $|x-3| < |x-5| \rightarrow x < 4$ by midpoint/distance reasoning.
- Line tangent to quadratic $\rightarrow$ eliminate a variable, set discriminant to zero.
- Remainder of $f(x)$ on division by $x-a \rightarrow f(a)$.

MM1. Algebra and functions - pattern bank I

High-value recognition patterns drawn from the official algebra, surd, quadratic, inequality and function material.

P1: Patterns That Should Fire Immediately

- Common base: rewrite powers before applying index laws. A power problem is often a base-recognition problem, not an arithmetic one.
- Surd denominator $a+b\sqrt{c}$: conjugate $a-b\sqrt{c}$. Difference of squares removes the surd from the denominator.
- $\sqrt{(A+B\sqrt{C})}$: compare with $(p+q\sqrt{C})^2$; use $2pq$ for the surd coefficient and p^2+Cq^2 for the rational part.
- Quadratic information: factorisation, formula, completed square, discriminant and graph are interchangeable views. Pick the one that exposes the requested information.
- Polynomial $f(x)$ with a suspected linear factor: test $f(a)=0$ before long division. The same substitution gives the remainder on division by $x-a$.

Ranked Speed Tricks

1. Factorise before machinery. Obvious roots and common factors can collapse a cubic or quartic before long division.
2. Use D as a structural test. $D>0$, $D=0$, $D<0$ gives two, repeated, or no real roots; after substitution this also gives two intersections, tangency, or no intersection.
3. Complete the square for graph questions. Vertex, translation, minimum/maximum and sometimes root existence appear immediately.
4. Keep surds exact. Decimalising destroys cancellation and may hide a conjugate or perfect-square structure.
5. In polynomial division, include missing powers as zero coefficients. This is a bookkeeping speed trick because it prevents alignment errors.

Common Wrong Turns

- Using the quadratic formula before checking factorisation; multiplying an inequality by an expression of unknown sign; treating $\sqrt{a+b}$ as $\sqrt{a}+\sqrt{b}$; forgetting that $\sqrt{x}$ denotes the non-negative root; assuming a few numerical substitutions prove an identity.

10-Second Check

- Can I factorise? Is there a conjugate? Would completing the square expose the answer? Does D answer the question without solving? Is the domain restricting a root?

MM1. Algebra and functions - pattern bank II

Inequalities, modulus, simultaneous equations and functions: preserve the information that ordinary equation manipulation can destroy.

Inequalities: Protect the Sign

You may add or subtract freely. Multiplication/division requires knowing the sign. For a rational inequality with denominator $g(x)$, multiplying by $g(x)^2$ can be useful because the square is non-negative (with $g(x)=0$ excluded from the domain). A number-line or graph method is often safer than splitting into many verbal cases.

Modulus and Distance Shortcuts

For weighted distances such as $2|x-a| < |x-b|$, sketch the two V-shaped graphs or solve branch intersections; the geometry guides which signs apply.

Do not square blindly unless you have checked equivalence. Squaring is safe for comparisons of known non-negative quantities, but can introduce or hide conditions in more general manipulations.

Simultaneous Equations and Functions

Line + quadratic: substitute the line into the quadratic; one quadratic remains. Its discriminant gives the intersection count before roots are calculated.

Function discipline: a function includes its domain. Exactly one output is assigned to each permitted input; one-to-one is an additional property, not part of the definition.

Domain checks: square-root arguments must be non-negative in the real-number setting; denominators must be non-zero; logarithm restrictions appear later in MM5.

Identity versus equation: an identity is true for every permitted input. Prove by algebraic transformation, not by testing sample values.

Micro-Drill: Choose the Route

- $|x-3| < |x-5|$ -> distance interpretation, boundary $x=4$.
- $(2x+5)/(x+3) > 1$ -> avoid multiplying by $x+3$ without sign control; bring to one side or use a positive square.
- Line tangent to quadratic -> substitute, then set $D=0$.
- Claim "identity" -> transform one side; numerical checks are only checks.
- $\sqrt{x-2}$ present -> record $x \geq 2$ before solving.

10-Second Check

- Unknown-sign multiplier? Domain exclusion? Modulus as distance? Tangency via $D=0$? Is the statement an equation, identity or function-domain question?

MM2. Sequences and series

Official guide: pp. 47-62 | Sections MM2.1-MM2.4

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Make indexing and formula choice automatic. The calculations are rarely conceptually difficult; the lost time comes from off-by-one errors, prematurely assuming a recurrence cycle, or expanding far more of a binomial than the question needs.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	AP/GP recognition and sums	Know n th-term and sum formulae without derivation time.
AUTOMATE	Binomial target-term selection	Most questions need one coefficient/term, not a full expansion.
AUTOMATE	Indexing and term counts	p to q inclusive contains $q-p+1$ terms; this is a frequent source of silent errors.
FLUENT	Recurrence relations	Write enough terms to establish behaviour rather than guessing from a short pattern.
FLUENT	Derived/combined series	Factor or combine known series instead of starting from scratch.

MEMORISE COLD

- AP n th term: $u_n = a + (n-1)d$. AP sum: $S_n = n/2[2a + (n-1)d] = n(a+l)/2$.
- $1+2+\dots+n = n(n+1)/2$.
- GP n th term: $u_n = ar^{n-1}$. Finite GP: $S_n = a(1-r^n)/(1-r)$ for $r \neq 1$. Infinite GP: $S_{\text{infinity}} = a/(1-r)$ only for $|r| < 1$.
- From term p to q inclusive there are $q-p+1$ terms.
- Binomial: $(a+b)^n = \sum C(n,r)a^{n-r}b^r$; $C(n,r) = n!/[r!(n-r)!]$; $C(n,r) = C(n,n-r)$.
- $0! = 1$. In each binomial term, the exponents of the two bracket terms add to n .

RANKED SPEED TRICKS

1. Recognise AP by constant first difference; GP by constant ratio. Do not force a formula before establishing which structure you have.
2. For a chunk of a GP, factor out the first term of the chunk. $u_p + \dots + u_q = ar^{p-1}[1+r+\dots+r^{q-p}]$, a fresh GP with $q-p+1$ terms.
3. Alternative for chunks: subtract partial sums. $S_q - S_{p-1}$ is often even faster if S_n is already available.
4. Recurrences: write enough terms to prove the cycle. The official notes explicitly warn against declaring repetition after too few terms.
5. Binomial: solve the exponent requirement first. Work out which r gives the required power of x , then calculate only that one term.
6. Raise the entire bracket term, coefficient and sign included. In $(2x-3)^n$, the contribution is $(2x)^{n-r}(-3)^r$, not $2x^{n-r}3^r$.
7. Use coefficient symmetry. $C(n,r) = C(n,n-r)$ can turn an awkward factorial calculation into a small one.
8. Do not expand a series term-by-term if a linear combination is obvious. Adding APs termwise produces another AP; exploit first terms and common differences directly.
9. For infinite GP, test $|r| < 1$ before using the formula. If it fails, there is no finite sum to infinity.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using n rather than $n-1$ in an n th-term exponent or AP term.
- Counting $q-p$ terms rather than $q-p+1$.
- Using S_{infinity} when $|r| \geq 1$.
- Expanding a full binomial to find one coefficient.
- Dropping the minus sign in a powered term such as $(-3)^r$.
- Assuming a recurrence pattern from two or three suggestive terms.

UNDER TIME PRESSURE: 10-SECOND CHECK

- AP or GP?
- Term number or number of terms?
- $q-p+1$?
- Need full expansion or one binomial term?
- $|r| < 1$?
- Recurrence cycle genuinely repeated?

MM3. Coordinate geometry in the (x, y) plane

Official guide: pp. 63-78 | Sections MM3.1-MM3.3

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Switch instantly between equation and picture. Lines, circles and circle theorems are most powerful when algebraic and geometric information are used together rather than treated as separate topics.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Lines and gradients	Point-gradient form and parallel/perpendicular conditions should be immediate.
AUTOMATE	Circle equation and completing square	Centre/radius must be read or recovered quickly.
AUTOMATE	Tangency and intersections	Substitution plus discriminant zero is the universal analytic shortcut.
FLUENT	Circle theorems	Often replace coordinate algebra with a short angle/length deduction.
FLUENT	Distance and translation ideas	Moving the picture mentally can simplify line-circle distance questions.

MEMORISE COLD

- Line through (x_1, y_1) with gradient m : $y - y_1 = m(x - x_1)$. General form $ax + by + c = 0$.
- Parallel non-vertical lines: equal gradients. Perpendicular gradients: $m_1 * m_2 = -1$.
- Circle centre (a, b) , radius r : $(x - a)^2 + (y - b)^2 = r^2$.
- For $x^2 + y^2 + dx + ey + f = 0$, complete squares separately in x and y ; centre is $(-d/2, -e/2)$, with radius determined by the remaining constant.
- Circle properties: perpendicular from centre to chord bisects it; tangent is perpendicular to radius at contact; centre angle is twice circumference angle on same arc; angle in semicircle 90 degrees; same-segment angles equal; opposite cyclic angles sum 180 degrees; alternate segment theorem.

RANKED SPEED TRICKS

- Use point-gradient form first. It avoids solving for an intercept before you need it.
- Perpendicular line: negative reciprocal immediately. Do not compute angles unless the question explicitly asks for them.
- Circle equation: complete x and y squares independently. Keep the added constants balanced on the same side before reading centre/radius.
- Tangency $\rightarrow$ discriminant zero. Substitute the line into the circle (or curve); exactly one intersection means repeated root.
- Use geometry before coordinates when a radius/tangent/chord theorem is visible. A right angle or isosceles triangle can remove several lines of algebra.
- Distance from a centre to a line is the controlling quantity in many circle problems. For shortest distance to the circumference, compare centre-line distance with r .
- Translate the picture mentally if it simplifies coordinates. A circle centred at (a, b) behaves like an origin-centred circle after shifting coordinates.

8. Read vertical/horizontal lines correctly. $x=\text{constant}$ is vertical; $y=\text{constant}$ is horizontal. Avoid trying to assign a finite gradient to a vertical line.

ADDITIONAL SPEED TRICKS — *new, not in the original guide*

1. Two chords, one centre: the perpendicular bisectors of any two chords of a circle intersect at the centre. A fast, reliable route into "find the centre given three points on a circle" style unfamiliar-application problems — exactly the kind the TMUA favours.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Reading r^2 as r .
- Forgetting the sign change in the centre: $(x-a)^2$ means centre $x=a$.
- Using $m_1*m_2=-1$ when one line is vertical; handle vertical/horizontal directly.
- Solving a tangent problem fully when discriminant zero gives the required parameter.
- Ignoring a simple circle theorem and doing coordinate algebra instead.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Line form?
- Parallel/perpendicular?
- Circle already completed square?
- Tangency $\rightarrow D=0$?
- Circle theorem shortcut?
- Radius or radius squared?

MM4. Trigonometry

Official guide: pp. 79-98 | Sections MM4.1-MM4.6

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Choose the right triangle tool, then control all possible angles. The mechanics are standard; the ESAT-level difficulty lies in ambiguous sine-rule cases, radian fluency, graph symmetries and not losing solutions when the argument is transformed.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Exact values and graphs	These should be recall, not calculation.
AUTOMATE	Trig equations over intervals	Finding every solution is the central trap.
AUTOMATE	Sine/cosine rule selection	The wrong rule creates needless algebra or ambiguity.
AUTOMATE	Ambiguous sine-rule case	SSA can produce two valid triangles.
FLUENT	Radians, arcs and sectors	Straightforward if pi-based angle conversions are automatic.
FLUENT	Identities	$\tan = \sin/\cos$ and $\sin^2 + \cos^2 = 1$ convert mixed equations into one function.

MEMORISE COLD

- Area of triangle = $(1/2)ab \sin C$ for included angle C.
- Sine rule: $a/\sin A = b/\sin B = c/\sin C$. Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$.
- 360 degrees = 2π radians. Arc length $s = r \theta$; sector area = $(1/2)r^2 \theta$ when θ is in radians. Segment area = sector area - triangle area.
- Exact values for 0, 30, 45, 60, 90 degrees. Derive from 45-45-90 and 30-60-90 triangles if recall fails.
- Periods: $\sin/\cos$ 2π (360 degrees); $\tan$ π (180 degrees).
- $\tan \theta = \sin \theta / \cos \theta$; $\sin^2 \theta + \cos^2 \theta = 1$.

RANKED SPEED TRICKS

1. Rule selection by data pattern. Right triangle $\rightarrow$ Pythagoras/basic trig first. Three sides $\rightarrow$ cosine rule. Two sides + included angle $\rightarrow$ cosine rule or area. Matched side-angle pairs $\rightarrow$ sine rule.
2. For an angle from three sides, prefer cosine rule. It avoids the sine-rule ambiguous case because cosine is one-to-one on 0 to 180 degrees.
3. SSA via sine rule: always check the supplementary angle. If $\sin B = s$, both B and $180-B$ may fit; reject only by triangle geometry.
4. Radians: work in π fractions. Memorise $\pi/6, \pi/4, \pi/3, \pi/2, \pi, 2\pi$ rather than converting numerically.
5. Segment area: sector minus triangle. Do not memorise it as a mysterious standalone formula; this prevents sign/major-segment errors.
6. Trig equation with transformed argument: solve for the argument over its full induced range first. Only then undo the transformation. The official notes explicitly warn that isolating x too early can lose solutions.
7. Mixed $\sin/\cos$ quadratic: use $\sin^2 + \cos^2 = 1$. Convert to one trig function, substitute a letter if helpful, then solve the quadratic.
8. Use graph periodicity/symmetry as a solution-count check. If your number of answers is inconsistent with the interval and period, recheck before finalising.

9. Use unit-circle/projection thinking for signs. It is more robust than memorising isolated quadrant mnemonics.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using sine rule without considering the ambiguous SSA case.
- Measuring radians as though they were degrees in arc/sector formulae.
- Losing solutions by isolating x before generating all solutions for the transformed argument.
- Forgetting $\tan$ is undefined where $\cos=0$.
- Using the wrong side opposite the named angle in sine/cosine rules.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Right triangle?
- Which rule matches the data?
- SSA ambiguity?
- Degrees or radians?
- All solutions in the full argument range?
- Use identity to reduce to one trig function?
- Solution count plausible from period?

EXPANDED PATTERN BANK

MM4. Trigonometry - solution discipline

The calculations are standard. The difficult part is retaining every valid angle and using the correct range.

P1: Solve the Argument First

For $\sin(kx+c)=a$, $\cos(kx+c)=a$ or $\tan(kx+c)=a$, set $u=kx+c$. Convert the stated x -range into the corresponding u -range. Find all u -solutions in that full range using periodicity and symmetry. Only then solve back for x . This prevents the classic lost-solution error.

Exact-Value and Sign Bank

Memorise/derive: 30, 45, 60 degrees from a 30-60-90 triangle and a 45-45-90 triangle. Keep values exact as $1/2$, $\sqrt{2}/2$, $\sqrt{3}/2$, $1/\sqrt{3}$, 1 , $\sqrt{3}$.

Signs: use the unit-circle/projection picture. $\cos$ is the x -coordinate, $\sin$ the y -coordinate, $\tan=\sin/\cos$. This is more robust than isolated sign mnemonics.

Periods: $\sin/\cos$ 2π (360 degrees); $\tan$ π (180 degrees). Use the period to estimate how many solutions should appear in the interval.

Mixed Trig Expressions

Quadratic-looking expression: use $\sin^2+\cos^2=1$ to reduce to one function, then substitute $s=\sin(\theta)$ or $c=\cos(\theta)$. Solve the ordinary quadratic, reject values outside $[-1,1]$, then solve the remaining trig equations over the full range.

$\tan$: remember $\tan=\sin/\cos$ and is undefined where $\cos=0$; do not create an apparent solution at a forbidden point by multiplying through by $\cos$ without checking.

Micro-Drill

- $\sin(2x+30 \text{ degrees})=1/2$ on $0 \leq x < 360 \rightarrow$ solve $u=2x+30$ on $30 \leq u < 750$ first.
- $12\cos^2(\theta)+6\sin(\theta)-12=0 \rightarrow$ replace $\cos^2$ by $1-\sin^2$; solve a quadratic in $\sin(\theta)$.
- $\tan(\theta)=-1/\sqrt{3} \rightarrow$ basic angle $\pi/6$, then use $\tan$ period π and the requested range.

10-Second Check

- Degrees or radians? What is the argument range? Have I generated every periodic/symmetric solution before rearranging? Any tan undefined points? Any algebraic root outside $[-1,1]$?

MM4. Trigonometry - triangles, radians and sectors

Rule selection should be driven by the data pattern, not by whichever formula is easiest to remember.

Triangle Tool Selection

- Right angle present: Pythagoras/basic trig first.
- Three sides known, angle wanted: cosine rule.
- Two sides and included angle, third side wanted: cosine rule.
- Two sides and included angle, area wanted: $(1/2)ab \sin C$.
- Known side-angle pair plus another side/angle: sine rule.
- SSA using sine rule: check the supplementary angle; there may be two valid triangles.

Radian Fluency

Memorise $30=\pi/6$, $45=\pi/4$, $60=\pi/3$, $90=\pi/2$, $180=\pi$, $360=2\pi$. Work with π -fractions rather than decimal radians. For θ in radians: arc length $s=r\theta$ and sector area $A=(1/2)r^2\theta$. A circular segment is sector minus the triangle formed by the two radii and chord.

Ranked Speed Tricks

1. Prefer cosine rule for an angle from three sides. It avoids the sine-rule ambiguous case because cosine is one-to-one on 0 to 180 degrees.
2. Draw the opposite side-angle pairing before using sine rule. Most formula errors are actually labelling errors.
3. Major/minor segment: identify which central angle the question uses before applying sector-minus-triangle.
4. 3D triangle questions are still 2D. Identify the relevant triangle in the solid, derive any hidden length with Pythagoras, then use the ordinary rule.

Ambiguous-Case Check

If $\sin B=s$ gives an acute B , also test $180 \text{ degrees}-B$. Keep it only if the resulting triangle has positive angles summing to 180 degrees and satisfies the stated geometry. Do not discard the second angle merely because a calculator first returns the acute one.

10-Second Check

- Right triangle? Which data pattern? Matched opposite pairs? SSA ambiguity? Radians? Sector or segment? Is the central angle major or minor?

MM5. Exponentials and logarithms

Official guide: pp. 99-107 | Sections MM5.1-MM5.3

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Treat logarithms as inverse exponents and preserve domain information. The syllabus is compact; speed comes from seeing a common base or a hidden quadratic in a^x before doing anything else.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Log laws and inverse relationship	These must be instant to keep exact solutions short.
AUTOMATE	Hidden quadratic in a^x	A substitution can turn an exponential equation into an ordinary quadratic.
AUTOMATE	Domains	Log arguments must be positive; this can remove algebraic roots.
FLUENT	Exponential/log graph recognition	Shape, intercepts and inverse relationship provide rapid checks.

MEMORISE COLD

- $a^x = b$ iff $x = \log_a b$.
- $\log_a x + \log_a y = \log_a(xy)$; $\log_a x - \log_a y = \log_a(x/y)$; $k \log_a x = \log_a(x^k)$.
- $\log_a a = 1$; $\log_a 1 = 0$. Base $a > 0$, $a \neq 1$; log argument must be positive.
- Graph $y = a^x$ passes through $(0, 1)$; $y = \log_a x$ is its inverse/reflection in $y = x$ and passes through $(1, 0)$.

RANKED SPEED TRICKS

- Find a common base before taking logs. $32^x = 4$ is easier when both are powers of 2.
- Spot the quadratic substitution. If you see a^{2x} , a^x and constants, let $y = a^x$; then $a^{2x} = y^2$.
- Combine logs before solving. A sum/difference of logs usually becomes one product/quotient inside a single log.
- Keep exact answers as logarithms. There is no reason to approximate unless specifically requested.
- Domain check after algebra. A root that makes any original log argument ≤ 0 is invalid.
- Graph check: exponential output is positive. If an intermediate solution implies $a^x \leq 0$ for positive a , something has gone wrong.
- Choose the log base for convenience. Any valid base works algebraically; a matching base can make the result cleaner.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Applying $\log(x+y) = \log x + \log y$. False.
- Forgetting log arguments must be positive.
- Missing the substitution $y = a^x$ in a quadratic-looking exponential equation.
- Rounding exact logarithmic answers unnecessarily.
- Confusing the domain of log with the range of log; log output itself can be negative.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Common base?
- Hidden quadratic in a^x ?
- Can logs combine?
- All arguments >0 ?
- Exact log form preferable?
- Graph positivity check?

MM6. Differentiation

Official guide: pp. 108-117 | Sections MM6.1-MM6.3

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

The calculus technique is deliberately narrow. The speed skill is to rewrite the expression as simple powers before differentiating, then interpret the derivative correctly as gradient/rate information.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Simplify before differentiating	No product/quotient/chain-rule machinery is required; algebraic rewriting is the intended route.
AUTOMATE	Stationary points and monotonicity	Derivative sign converts directly into shape information.
AUTOMATE	Tangents and normals	One derivative value plus point-gradient form should be automatic.
FLUENT	Second derivative classification	Fast for maxima/minima; remember zero is inconclusive.
FLUENT	Rate-of-change interpretation	Translate physical/geometric wording into derivative meaning.

MEMORISE COLD

- $d/dx(x^n) = n x^{n-1}$ for rational n ; differentiate sums term by term; derivative of a constant is zero.
- Gradient of tangent at $x=a$ is $f'(a)$. Normal gradient is $-1/f'(a)$ when the tangent gradient is finite and non-zero.
- Stationary points satisfy $f'(x)=0$.
- At a stationary point, $f''(x)>0$ gives local minimum; $f''(x)<0$ gives local maximum; $f''(x)=0$ is inconclusive.
- If $f'(x)>0$ on an interval the function is strictly increasing there; if $f'(x)<0$ it is strictly decreasing. The official notes warn not to reverse these implications mechanically in subtle cases.
- Points of inflexion are expected only qualitatively, not as a technical differentiation test.

RANKED SPEED TRICKS

1. Rewrite first, differentiate second. Expand brackets, split algebraic fractions and turn roots into fractional powers before applying the power rule.
2. Tangent/normal: calculate only what is needed. Find the point, one derivative value, then use point-gradient line form.
3. Stationary points: factor the derivative if possible. Root-finding on f' is usually shorter than expanding further.
4. Classify using the quickest valid evidence. Second derivative sign is fast; derivative sign change is more general; known curve shape can also guide.
5. Use derivative sign as a graph tool. Intervals of + and - give increasing/decreasing behaviour without plotting many points.
6. Remember derivative is a rate. Units of dy/dx are y-units per x-unit; this is a useful interpretation check.
7. Do not overcomplicate inflexions. The official scope asks only qualitative understanding; focus on maxima/minima and monotonicity.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Trying to use chain/product/quotient-rule machinery instead of simplifying an expression that the syllabus intends you to rewrite.
- Forgetting the negative reciprocal for a normal.
- Assuming $f''=0$ proves an inflexion or rules out a max/min.
- Reversing a sufficient condition: $f'>0$ guarantees strictly increasing, but a strictly increasing function can have isolated $f'=0$ points.
- Solving $f(x)=0$ instead of $f'(x)=0$ for stationary points.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Simplified to powers?
- Tangent or normal?
- Stationary $\rightarrow f'=0$?
- Need classification?
- Derivative sign intervals?
- Any logical implication being reversed?

MM7. Integration

Official guide: pp. 118-135 | Sections MM7.1-MM7.6

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Keep signed integrals distinct from geometric area. The integration rules are simple; most traps concern limits, signs, symmetry and forgetting the constant in indefinite integration.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Signed integral versus area	A curve below the axis contributes negative integral but positive geometric area.
AUTOMATE	Power rule and simplification	As with differentiation, algebraic rewriting is the intended speed route.
AUTOMATE	Fundamental Theorem and limits	Evaluation and variable-limit differentiation should be immediate.
AUTOMATE	Symmetry and combining ranges	Can remove nearly all computation.
FLUENT	Trapezium rule	Mechanically easy; the interpretation of over/under-estimate matters.
FLUENT	Simple differential equations	Integrate and use the supplied point to find the constant.

MEMORISE COLD

- Integral of $x^n dx = x^{n+1}/(n+1) + C$ for $n \neq -1$.
- Definite integral: $\int_a^b f(x)dx = F(b) - F(a)$, where $F' = f$.
- $d/dx [\int_a^x f(t)dt] = f(x)$.
- Swap limits $\rightarrow$ change sign. Contiguous ranges combine; equal ranges can combine integrands.
- Even function over $[-a, a]$: integral = $2 \int_0^a$. Odd function over $[-a, a]$: integral = 0.
- Trapezium rule with equal step h : $h/2 [y_0 + 2y_1 + \dots + 2y_{n-1} + y_n]$.
- Solving $dy/dx = f(x)$: integrate to get $y = \int f(x)dx + C$, then use a point to determine C .

RANKED SPEED TRICKS

1. Ask "integral or area?" before calculating. If the curve crosses the x-axis and geometric area is requested, split at roots and add absolute areas.
2. Simplify to powers first. Expand/split/rewrite roots exactly as in differentiation.
3. Exploit symmetry before antiderivatives. Odd integrand on symmetric limits gives zero instantly; even integrand halves the work.
4. Combine ranges algebraically before evaluating. Adjacent integrals can collapse into one known interval; reversed limits introduce a minus sign.
5. Fixed numerical limits mean a constant. The derivative of a definite integral with two fixed endpoints is zero; the FTC derivative rule needs a variable limit.
6. Trapezium over/under: sketch concavity. Straight chords lie above a convex-up curve and below a concave-down curve; use the picture rather than a memorised slogan if unsure.
7. Differential equation with one point: choose indefinite or definite setup. Both are valid; use whichever gives fewer constants/steps.

8. Never distribute integration over products/quotients as products/quotients of integrals. Linearity applies to sums and constant multiples, not arbitrary multiplication or division.

ADDITIONAL SPEED TRICKS — *new, not in the original guide*

1. The odd part still vanishes inside a mixed polynomial: split any polynomial into its odd-degree and even-degree terms mentally. Over symmetric limits, the odd-degree part integrates to zero regardless of what else is mixed in — you only need to integrate the even part. More general than only spotting a pure odd/even function.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Forgetting $+C$ in an indefinite integral.
- Reporting a negative definite integral as a negative geometric area.
- Failing to split at an x-axis crossing when area is required.
- Dropping the minus sign after reversing limits.
- Using the trapezium formula with the wrong number of ordinates or step size.
- Treating $\text{integral}(fg)$ as $\text{integral}(f) \text{integral}(g)$.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Signed integral or positive area?
- Crosses axis?
- Symmetry?
- Can ranges combine?
- Step size correct?
- $+C$?
- Variable limit or fixed limits?

MM8. Graphs of functions

Official guide: pp. 136-159 | Sections MM8.1-MM8.7

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Use graphs as a compressed representation of algebra. Transformations, roots, stationary points and intersections should let you predict shape and solution structure before doing detailed calculations.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Graph transformations	Inside versus outside changes, signs and order of composition are high-frequency reasoning tools.
AUTOMATE	Standard graph shapes	A sketch should start from known families, not a table of arbitrary values.
AUTOMATE	Roots/intersections as equations	Geometric interpretation can turn an awkward algebraic question into a simple counting/position question.
AUTOMATE	Differentiation-assisted sketching	Stationary points and monotonicity complete unfamiliar polynomial sketches.
FLUENT	Parameter effects in lines/quadratics	Read slope, intercept, translation and stretch directly from form.
FLUENT	Function composition	Substitute the whole inner expression and remember order generally matters.

MEMORISE COLD

- $y=a f(x)$: vertical scale by a ; if $a<0$ also reflect in x -axis.
- $y=f(x)+a$: vertical translation by $(0,a)$.
- $y=f(x+a)$: horizontal translation by $(-a,0)$.
- $y=f(ax)$: horizontal scale factor $1/|a|$; if $a<0$ also reflect in y -axis.
- $y=|f(x)|$: reflect the parts below the x -axis upwards.
- $f(g(x))$ means substitute $g(x)$ into every occurrence of x in f ; generally $f(g(x)) \neq g(f(x))$.
- Roots of $f(x)=0$ are x -axis intersections. Solutions of $f(x)=g(x)$ are x -coordinates of intersections of the two graphs.
- For an accurate unfamiliar polynomial sketch: intercepts/roots, stationary points, increasing/decreasing intervals and end behaviour.

RANKED SPEED TRICKS

1. Transform inside: factor first. For $f(2x+c)$, rewrite as $f(2(x+c/2))$ so the horizontal scale and translation are visible. The official notes explicitly warn that naive reading of the raw constant gives the wrong shift.
2. Inside changes feel backwards. $f(x+a)$ shifts left by a ; $f(ax)$ compresses horizontally for $|a|>1$.
3. Apply transformations in the order encoded by the algebra. Compositions do not always commute, particularly when two changes act on the same axis.
4. Complete the square to sketch a quadratic. It immediately exposes horizontal/vertical shift and turning point.
5. Use factorisation for x -intercepts, differentiation for turning behaviour. These two tools usually determine a polynomial sketch with little point plotting.

6. Before solving an ugly equation, ask if it is an intersection problem. Sketching f and g may reveal number/location of solutions or eliminate options.
7. End behaviour from leading term. Degree parity and sign of the leading coefficient tell you what happens as $x \rightarrow \pm \infty$.
8. For modulus graphs, sketch first then flip only the negative y -parts. Do not reflect the whole graph.
9. Parameter questions: isolate the role of each parameter. In $y=mx+c$, m controls slope and c vertical intercept; in $a(x+b)^2+c$, a controls vertical scale/reflection, $-b$ the vertex x -coordinate, c the vertex y -coordinate.

OPTIONAL EXTRAS — *useful if already familiar, not a preparation priority*

- $f(|x|)$ vs $|f(x)|$, as a pair: $|f(x)|$ reflects any negative- y part of the graph up above the x -axis; $f(|x|)$ instead discards the $x < 0$ branch entirely and mirrors the $x \geq 0$ branch onto it. Useful to have straight if you already know it; not a preparation priority.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Shifting $f(x+a)$ to the right instead of left.
- Using horizontal scale factor a rather than $1/a$ for $f(ax)$.
- Applying a translation before factoring an internal expression such as $2x+c$.
- Reflecting all of f for $|f|$ rather than only negative portions.
- Plotting many points before using roots/stationary points/end behaviour.
- Assuming $f(g(x))=g(f(x))$.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Known base graph?
- Inside or outside transformation?
- Factor internal linear argument?
- Roots/intersections?
- Stationary points?
- End behaviour?
- Modulus only flips negative y ?

EXPANDED PATTERN BANK

MM8. Graphs of functions - transformations

Treat a transformed expression as instructions applied to coordinates. Inside changes act horizontally and appear reversed.

Memorise Cold

- $y=a f(x)$: vertical scale by a ; if $a < 0$, reflect in x -axis.
- $y=f(x)+a$: translate by $(0,a)$.
- $y=f(x+a)$: translate by $(-a,0)$.
- $y=f(ax)$: horizontal scale factor $1/|a|$; if $a < 0$, reflect in y -axis.
- $y=|f(x)|$: leave non-negative parts; reflect only the parts below the x -axis upwards.
- $f(g(x))$: substitute the entire expression $g(x)$ into every occurrence of x in f .

Ranked Speed Tricks

1. Factor inside before reading the shift. $f(2x+c)=f(2(x+c/2))$; the horizontal shift is $-c/2$, not $-c$.
2. Apply transformations in the algebraic order. Operations on the same axis do not generally commute.

3. Track landmark points. Roots, intercepts, turning points and asymptotic behaviour are faster to transform than a large table of points.
4. Complete the square before transforming a quadratic. $a(x-h)^2+k$ exposes vertex, reflection and vertical scale.
5. Negative scale factor = scale plus reflection. Separate those two ideas mentally if the combined rule feels opaque.

Micro-Drill

- $f(x+3) \rightarrow$ left 3.
- $f(2x) \rightarrow$ horizontal scale factor $1/2$.
- $-2f(x) \rightarrow$ vertical stretch factor 2 and reflection in x-axis.
- $f(-x) \rightarrow$ reflection in y-axis.
- $|f(x)| \rightarrow$ only negative y-parts move.
- $\cos(2x+\pi/6)=\cos(2(x+\pi/12)) \rightarrow$ horizontal scale $1/2$, then left $\pi/12$.

10-Second Check

- Inside or outside? Have I factored the inside? Which axis? Does a negative factor also reflect? Can I transform key points instead of plotting afresh?

MM8. Graphs of functions - sketching and intersections

A graph is a compressed statement about roots, signs, monotonicity and simultaneous solutions.

Full Sketch Checklist

1. End behaviour: degree/parity and leading coefficient for polynomials.
2. x-intercepts: factorise or otherwise locate roots; note repeated roots because they touch rather than cross.
3. y-intercept: evaluate $f(0)$ when defined.
4. Stationary points: solve $f'(x)=0$ and classify if needed.
5. Increasing/decreasing intervals: use derivative sign where useful.
6. Domain/asymptotic restrictions: especially reciprocal, root and logarithmic families.

Intersections as a Speed Tool

$f(x)=0 \rightarrow$ x-axis intersections.

$f(x)=g(x) \rightarrow$ intersections of $y=f(x)$ and $y=g(x)$. A sketch may reveal the number and approximate location of solutions without exact algebra.

Tangency $\rightarrow$ one repeated intersection; algebraically a discriminant-zero condition after substitution, geometrically a touching point.

Parameter questions $\rightarrow$ ask how changing the parameter moves, stretches or rotates the graph. The critical value often occurs when two roots merge or a graph just touches another.

Standard Families to Recognise on Sight

Linear; quadratic; cubic and higher polynomials; reciprocal $1/x$; exponential a^x ; logarithm $\log_a x$; square root; modulus; sin, cos and tan. The goal is not detailed plotting from scratch: identify symmetry, domain, intercepts, asymptotes/undefined points and end behaviour first.

Common Wrong Turns

- Plotting many arbitrary points before using structure; forgetting repeated roots touch the axis; translating $f(x+a)$ to the right; applying the raw constant in $f(2x+c)$ before factoring; assuming $f(g(x))=g(f(x))$; reflecting the entire graph for $|f(x)|$.

10-Second Check

- What family? Roots/intercepts? Repeated root? End behaviour? Stationary points? Domain? Could the equation be answered as an intersection-count question rather than solved exactly?

Final recall sheet

Final high-speed recall sheet

Highest-value habits

- Factorise and simplify before using general machinery.
- Use discriminants for root counts and tangency, not only for solving quadratics.
- Treat modulus as distance and graph transformations as coordinate changes.
- AP/GP indexing: n th term uses $n-1$; p to q inclusive has $q-p+1$ terms.
- Binomial questions usually need one term, not the full expansion.
- Tangency of line and curve $\rightarrow$ substitute, discriminant zero.
- SSA sine rule $\rightarrow$ consider the supplementary angle.
- Trig equations: solve for the transformed argument across its full range first.
- Logs: combine, check domain, spot hidden quadratic in a^x .
- Differentiate/integrate only after rewriting into simple powers.
- Definite integral is signed; geometric area is positive.
- Use symmetry and graph structure whenever they remove arithmetic.

How to practise this document

- Pass 1: cover the green boxes and reproduce them from memory.
- Pass 2: read only the ranked speed tricks and say when each shortcut is valid.
- Pass 3: do timed official questions and record whether errors were knowledge, method selection, algebra/arithmetic, or interpretation.
- Pass 4: revisit only the unit that generated the error; do not reread everything.